

Aluminium–Lithium (Al–Li) Alloys: Executive Summary

Al–Li alloys are advanced high-strength, low-density aluminium alloys engineered for aerospace (wing and fuselage skins, cryogenic fuel tanks), high-performance transport, and sports applications. By alloying aluminium with ~1–3 wt% Li (the lightest metal), density is reduced by ~3% per 1 wt% Li and elastic modulus increases by ~5–6% [48†L143-L151] . Typical 2nd-generation Al–Li alloys (e.g. AA2090, AA2091, AA8090) contain ~2–2.5% Li and 1–3% Cu/Mg, yielding ~8–10% weight savings vs 2xxx/7xxx series at similar strength [35†L201-L204] [48†L143-L151] . Third-generation alloys (e.g. AA2195, AA2198, AA2099, Weldalite™ 049) use <2% Li with higher Cu/Li ratios to restore toughness [73†L209-L217] [63†L532-L538] . Al–Li alloys reach tensile strengths comparable to high-strength Al–Cu and Al–Zn alloys (500–600 MPa) but at ~10% lower density (≈ 2.6 – 2.7 g/cc vs 2.78 – 2.81 for 2024-T3/7075-T6) [63†L521-L526] [41†L52-L60] . Elastic moduli (~69–78 GPa) are slightly higher than standard alloys, and fatigue/damage tolerance are excellent under tension [41†L52-L60] [63†L540-L544] . However, Al–Li alloys are more anisotropic (rolling texture), have lower fracture toughness (often ~25–35 MPa $\sqrt{m}$) and poorer corrosion resistance than conventional Al alloys [69†L214-L219] . Processing costs are high: raw material costs are several times those of 2xxx/7xxx alloys and manufacturing (strict melting control, special heat treatments, FSW) adds further expense [69†L214-L219] .

Key merits: weight and stiffness improvements drive fuel savings ($\approx 1\%$ fuel/1% weight) [63†L568-L572] . Many modern aircraft (Airbus A350, Boeing 777X) use 3rd-gen Al–Li for fuselage skins, enabling ~6–9% airframe weight reduction [63†L521-L529] . **Limitations:** high cost (~3 $\times$ conventional Al [69†L214-L219]), sensitivity to heat/water (Li is reactive), and tighter process control (Li precipitates, PFZs) require special handling. Careful design (e.g. avoid highly strained compression panels, use tension skins) and protective coatings are mandatory. Additive-manufacturing of Al–Li is under development. In sum, Al–Li alloys offer unique strength-to-weight advantages for critical aerospace structures but at significant cost and processing complexity. This report reviews their metallurgy, properties, processing, applications, cost/benefit, and best practices for design and manufacture.

Alloy Chemistry and Commercial Grades

Al–Li alloys are classified by **generation**: first (1940s–60s, low Li ~1–2%), second (1970s–90s, Li $\geq 2\%$), and third (1990s–present, Li<2%). Key aerospace Al–Li alloys (all in 2000-7000 series family) and typical composition ranges are:

Alloy (AA)	Li (wt%)	Cu (wt%)	Mg (wt%)	Ag (wt%)	Zr (wt%)	Density (g/cm ³)	Applications
2090 (2nd gen)	1.90–2.60	2.40–3.00	≤ 0.25	–	≤ 0.14 (Zr)	~2.59 [28†L172-L180]	Fuselage/wing skins (e.g. A330/ A340 leading edges), C-17, Atlas rockets

Alloy (AA)	Li (wt%)	Cu (wt%)	Mg (wt%)	Ag (wt%)	Zr (wt%)	Density (g/cm ³)	Applications
2091 (2nd gen)	~2.0	~2.0	~1.3	–	0.11 (Zr)	~2.58 【35†L200-L204】	Fuselage fairing, access panels
8090 (2nd gen)	~2.4	~1.2	~0.8	–	0.11 (Zr)	~2.54 【35†L201-L204】	Fuselage skins (e.g. A330/A340), empennage
2195 (3rd gen)	0.8–1.2	3.7–4.3	0.25–0.8	0.25–0.6	0.08–0.16 (Zr)	~2.70 【73†L209-L217】	Cryogenic tanks (Shuttle external tank), high-speed aircraft
2198 (3rd gen)	~1.0	~3.2	~0.5	~0.4	0.11 (Zr)	2.69 【73†L220-L223】	Fuselage skins (Airbus A350, A220), SpaceX Falcon 9
2099 (3rd gen)	~1.8	~2.7	~0.3	–	0.09 (Zr)	2.63 【73†L223-L227】	Lower fuselage skins (A350, 787/777X)
<i>Weldalite</i> [™] 049 (AA2095)	0.7–1.8	2.3–5.2	0.25–0.8	0.25–0.8	0.14 (Zr)	2.70 【54†L770-L777】	Ultra-high-strength, improved weldability
2050 (Al–Cu–Li–Ag, 3rd)	~1.0	3.6	0.4	0.4	0.11 (Zr)	~2.70 【73†L228-L233】	Wing/fuselage structure (A380, A350 ribs)

Table: Representative Al–Li alloys and composition. Zr (0.08–0.15%) is nearly constant in aerospace grades; Ag (silver) in third-gen (2195, 2196, 2050, Weldalite) promotes fine T1 precipitates for toughness 【73†L209-L217】 【63†L532-L538】 .

Commercial Al–Li alloys are typically wrought (rolled/extruded) alloys designated AA2xxx. Standards exist (e.g. AMS 4251/AMS 4303 cover 2090 sheet/plate; AMS 4179/4304 for 2198/2199, etc.) to define chemistry and mechanical spec. Table entries above cite compositional specs (e.g. AA2195: Li 0.8–1.2%, Cu 3.7–4.3% 【73†L209-L217】 ; Weldalite 049: Li 0.7–1.8%, Cu 2.3–5.2% 【54†L770-L777】).

Alloy design: Third-gen alloys lower Li (<2%) to reduce deleterious effects (PFZ, anisotropy, poor ductility) while retaining ~3–6% density saving. Alloy chemistry is optimized for high **strength/weight**: adding Li reduces density ~3% per 1% Li and raises stiffness ~5–6% 【48†L143-L151】 【63†L532-L538】 . Copper and magnesium supply classic Al–Cu–Mg hardening (forming S', T1, θ'), while Li adds Al₃Li (δ') coherent particles (L1₂ structure) that strengthen by modulus hardening 【63†L532-L538】 . Minor alloying (Ag, Zr, Sc, Mn) controls precipitate nucleation and grain structure: e.g. Ag coats T1 plates to refine them, Zr/Sc form fine Al₃Zr/Al₃(Sc,Zr) dispersoids to prevent grain growth and improve toughness 【63†L532-L538】 .

Microstructure and Metallurgy

Al-Li alloys age-harden via ordered coherent precipitates. The δ' (Al_3Li) phase (L1_2 cubic, coherent with Al) is the primary strengthening agent: a high volume fraction (~up to 20%) of δ' (Li content) significantly increases yield strength and stiffness [48†L143-L151] [63†L532-L538]. For example, each 1 wt% Li adds ~1.2 vol% δ' . T_1 (Al_2CuLi) is the dominant plate-like precipitate in Cu-bearing Al-Li-Cu alloys (third-gen), providing the highest strength. θ' (Al_2Cu) also forms if Li is low or Cu/Li <1. 2nd-gen alloys (high Li) can form coarse δ' and often an ordered AlLi (B32 β) phase if Li >~4%, but commercial grades keep Li <3% to avoid that.

Microstructural features:

- **Fine dispersoids (Al_3Zr)** from Zr additions create a stabilized grain structure (avoiding recrystallization) and enhance toughness [63†L532-L538]. Typical grain size is ~10–50 μm .
- **Precipitate-Free Zones (PFZs):** Li and Cu atoms diffuse to grain boundaries during heat treatment, creating Li-depleted zones next to boundaries that are weak in corrosion and fracture [3†L163-L168] [63†L532-L538]. This is a key weakness: Al-Li tends to be less corrosion-resistant (exfoliation, SCC) than Al-Cu alloys because PFZs act as initiation sites [69†L214-L219].
- **Anisotropy:** Cold-rolled Al-Li sheet develops strong crystallographic texture. Along the rolling direction, strength and ductility differ markedly from the transverse direction. Such anisotropy (up to ~20% variation in strength/toughness) requires careful forming design (multiple-crank panel forming, stretch forming) [69†L214-L219].

Heat treatment: Al-Li alloys use solution (≈ 520 – 550°C) and quench + artificial aging sequences. Common tempers: T8 (solution-treat + cold-work + age) for maximum strength (used in fuselage skin sheet like 2199-T8, 2195-T8); T6 (solution + age) for forgings; T81/T83 (solution + “overaging”) to improve toughness and reduce residual stresses (used for 2090 plate). Weldalite 049 (AA2095) is unique in being weldable by fusion methods (due to Ag/Mg additions) [54†L774-L782] [54†L788-L796], but is usually used in T8 condition (as “2095-T8”). Overaging (T7, T84, etc.) is sometimes employed for better SCC resistance at the expense of ~10% strength.

Mechanical Properties

Al-Li alloys offer a high **specific strength** (strength-to-weight) and high modulus for aluminium alloys. Table values for representative alloys (from datasheets and references) are:

Property	AA2024-T3	AA7075-T6	AA2199-T8E90	AA2099-T83	AA2195-T8	17-4PH (H900)	AISI 4340 Steel
Density (g/cm ³)	2.78	2.81	2.68 [63†L521-L526]	2.69 [63†L521-L526]	2.70 [73†L209-L217]	7.75	7.85
Yield Strength (MPa)	~290	~505	~380 [63†L521-L526]	~400 [63†L521-L526]	540 [63†L521-L526] (see note)	~1030	~1150

Property	AA2024-T3	AA7075-T6	AA2199-T8E90	AA2099-T83	AA2195-T8	17-4PH (H900)	AISI 4340 Steel
Ultimate UTS (MPa)	~440	~570	~430 【63†L521-L526】	~450 【63†L521-L526】	600 【63†L521-L526】	~1100	~1250
Elastic Modulus (GPa)	73	72	77 【63†L523-L526】	76 【63†L523-L526】	78 【63†L523-L526】	~200	~210
Elongation (%)	~19	~11	~6–7	~8–10	~5–9 【41†L24-L30】	~10	~10
Fracture Toughness, K_{IC} (MPa√m)	~20–30	~15–25	~30 【41†L36-L39】 (2195)	–	–	–	~50–100
Fatigue Strength (MPa, 10^7 cycles)	~150	~200	~190 【41†L32-L39】 (2195)	–	–	–	–
Key Use	Legacy airframe	High-str eng	3rd-gen fuselage skin	3rd-gen fuselage skin	Cryo tanks	Turbine disks, rotors	Landing gear, shafts

Data compiled from sources: AA2199/2195/2099 strengths from 【63†L521-L526】 ; AA2195 properties from MakeItFrom 【41†L24-L30】 【41†L34-L39】 ; steels from typical AMS specifications.

Interpretation: The table illustrates that 3rd-gen Al–Li alloys (e.g. AA2199, AA2099, AA2195) trade a few hundred MPa in strength for ~4–5% density reduction and ~5% higher modulus compared to legacy alloys. For example, 2199-T8E90 (2.68 g/cc) is ~3.6% lighter than 2024-T3 (2.78 g/cc) while its modulus is ~5% higher 【63†L523-L526】 . The ultimate example, AA2195-T8 (density 2.70 g/cc), can reach ~600 MPa UTS 【63†L523-L526】 —comparable to 7075-T6—coupled with excellent fatigue life 【41†L32-L39】 and stiffness (78 GPa). By contrast, steels have ~3× higher density: a 17-4PH or 4340 steel part is much heavier for the same strength (e.g. 4340: ~7.85 g/cc, YS ~1150 MPa). Thus Al–Li’s specific strength is far superior to steel.

Fatigue and Crack Growth: Al–Li alloys generally exhibit **high fatigue resistance** in tension; e.g. AA2195-T8 has an endurance limit ~190 MPa (for 2 mm unnotched) 【41†L32-L39】 . Fatigue crack growth rates in 3rd-gen alloys are comparable or better than 2xxx series; third-gen Al–Li is selected in fuselage skins partly for its damage tolerance 【63†L540-L544】 . However, under **compression loading** Al–Li alloys can suffer lower fatigue life (“fatigue in compression remains a problem” 【69†L214-L219】), necessitating design care.

Fracture Toughness: Typical fracture toughness of Al–Li is moderate. For example, AA2195-T8 has $K_{IC} \approx 30 \text{ MPa}\sqrt{\text{m}}$ 【41†L36-L39】 , similar to other high-strength Al–Cu alloys. This is lower than 2xxx alloys

(e.g. AA2024-T3 $\sim 25 \text{ MPa}\sqrt{\text{m}}$) and much lower than steels. Toughness tends to decrease with higher Li and Cu (large δ' volume, coarse T1). Third-gen alloys (lower Li) were developed to improve toughness.

Corrosion: Al-Li alloys generally corrode more easily than Al-Zn/Mg alloys due to PFZs and coarse intermetallics [69†L214-L219]. Exfoliation and pitting are concerns, and stress-corrosion cracking (SCC) thresholds are lower. In practice, parts are Alclad or coated; common aerospace treatments (chromate conversion, anodizing, epoxy primers) are applied. Some alloys (Weldalite 049) were engineered for better corrosion/SC resistance with Ag/Mg additions, but caution is always required in salt or acidic environments [54†L770-L777] [69†L214-L219].

Thermal Properties

Al-Li alloys have thermal properties similar to conventional aluminium:

- **Thermal conductivity:** High, $\sim 120\text{--}130 \text{ W/m}\cdot\text{K}$ [41†L82-L84] ($\approx 60\%$ of copper, comparable to 2xxx/7xxx Al). For example, AA2195: $130 \text{ W/m}\cdot\text{K}$ [41†L82-L84]. This aids machining and heat dissipation.
- **Thermal expansion:** $\sim 23 \mu\text{m/m}\cdot\text{K}$ [41†L84-L88] (room-temperature coefficient), close to other high-strength Al ($\sim 22\text{--}24 \mu\text{m/m}\cdot\text{K}$). This is much higher than steels, so thermal strain must be managed in bonded/joined structures.
- **Specific heat:** $\sim 900 \text{ J/kg}\cdot\text{K}$ [41†L78-L84] ($0.22 \text{ BTU/lb}\cdot^\circ\text{F}$), similar to pure Al.
- **Melting range:** Solidus $\sim 550^\circ\text{C}$, liquidus $\sim 660^\circ\text{C}$ [41†L68-L76] (dependent on Li content). These are slightly lower than pure Al (660°C), but still well above typical processing/working temperatures. Maximum recommended service temperature is usually $\sim 210^\circ\text{C}$ (above which creep may occur) [41†L64-L72].
- **Thermal stress and conductivity:** Li addition raises modulus, so Al-Li has slightly higher thermal stress under fixed constraints. Conductivity is lower in Li-rich alloys (Li scatters electrons), but remains high enough for heat exchange.

In short, thermal properties remain aluminium-like, with marginal changes due to Li. The key effect of Li on temperature-related behavior is often indirect (through processing/fatigue), not a dramatic change in thermal expansion or conductance.

Corrosion Behaviour and Surface Treatments

Al-Li alloys tend to have **poorer corrosion resistance** than Al-Zn or Al-Mg alloys. This is due to: (1) Li-rich precipitates and PFZs that are anodic; (2) an absence of protective cladding (Al-Li parts are rarely Alclad). Common forms of attack include pitting, exfoliation, and stress-corrosion cracking in saline or humid environments [69†L214-L219]. For example, 2nd-gen alloys (e.g. AA2090) can corrode rapidly if unprotected. The high copper content in many Al-Li alloys can exacerbate localized attack.

Mitigation: As with other Al airframe alloys, surfaces are treated. Typical treatments include trivalent chromate conversion coatings (e.g. Alodine), anodizing (thin oxide), and organic primers/paints. While Al-Li can be anodized, the resulting oxide may differ in color and adhesion; often a proprietary chem film or zirconium phosphate primer is used. Cladding with pure Al is **not** standard on Al-Li sheet (cladding would alter Li content and properties), so protective coating is critical. In practice, aerospace Al-Li parts receive strong corrosion protection (e.g. epoxy primer under paint, or Alodine + primer [69†L214-L219]).

Surface treatments: Solvent cleaning and etching must account for Li (avoid water traps). Fasteners and adhesives must be compatible with Al-Li chemistry. Plasma or chemical conversion coatings can enhance bond to adhesives and seal pores.

Overall, design for corrosion includes selecting appropriate alloys and tempers (e.g. slight overage, Cu/Zn levels), using sealants, and planning inspection/maintenance, especially for highly loaded or critical parts [69†L214-L219] .

Manufacturability

Al-Li alloys can be **cast, wrought, and fabricated** using many of the same methods as conventional aluminium, but with special precautions due to Li's reactivity and the alloys' mechanical anisotropy.

- **Casting:** Al-Li alloys are generally **poor for traditional casting** (sand/casting) because Li is highly reactive (ignite if moisture present) and tends to segregate (excessive Li can form brittle AlLi β phase). In practice, nearly all Al-Li commercial products are wrought (rolled/extruded) from ingot or billet cast under tightly controlled conditions (dry inert atmosphere, vacuum melting). On a few R&D efforts, near-net-shape castings (like spray-casting, powder metallurgy) have been explored, but there are no widespread cast Al-Li parts in service. Vacuum induction melting and fast solidification can minimize Li loss. In summary: commercial Al-Li starts from ingots poured in sealed, protective atmospheres, often degassed to remove H.
- **Hot Working (Rolling/Forging):** Wrought Al-Li is rolled into sheets (typically 1–20 mm for skins) or extruded/forged. As-cast ingots are homogenized and hot-rolled. Al-Li is generally **less forgeable** than 2xxx alloys at room temp, but can be worked at elevated T. Hot forging and hot rolling are routine. Arconic (formerly Alcoa) has produced plate up to 220 inches (5.6 m) wide [68†L1-L4] , enabling fewer welds in large fuselage panels. Hot rolling must avoid overheating (which coarsens δ') and excessive grain growth. After rough hot working, plates are cold-rolled to final gauge ($\geq 60\%$ reduction), then aged. The end grain shows elongated δ' plates. Cold-rolling is similar to Al 2024 but higher strength/low ductility can cause wrinkling or edge cracks if lubrication or anneal is inadequate. Multi-pass rolling with intermediate stress relief is often used.
- **Sheet Forming:** Forming (stretch forming, bending, deep drawing) of Al-Li sheet is similar to other 2xxx-series, but *higher springback* occurs due to high yield modulus ratio [37†L25-L28] . Designers must account for elastic recovery in dielines. Uniform deformation is difficult due to anisotropy – 3D-numerical forming simulations or specialized form tools (and multiple biaxial forming steps) are common. In practice, fuselage panels are stretch-formed over very large dies. Some processes (hydroforming, superplastic forming) have been explored for Al-Li sheet. Overall, forming requires good lubrication and gradual strain introduction to avoid cracks.
- **Extrusion:** Al-Li alloys can be extruded into bars, tubes, and profiles. Alloys like 2090, 2199 are offered in extruded shapes (AA2199 extrusions are available for fuselage frames). Extrusion ratios (area reduction) are comparable to 2xxx alloys; proper lubrication and pre-heating (around 430–450°C) give high quality. Tooling must be hard-faced to withstand Al-Li's high strength. Quenching is typically water or polymer-based. Post-extrusion aging yields T8 or T6 tempers as required.
- **Heat Treatment:** After fabrication (rolling, forging, extruding), alloy is solution-treated (T4) and quenched (usually cold-water). Due to Al-Li's high strength, some tempers use *cold work + age*

(T8/T81) instead of or in addition to T6 for extra yield. Temper T8 (ex. 2090-T81, 2199-T8) yields maximum strength: it is solution-treated, cold-worked (~5–8% stretch), then aged. Overaged tempers (T7, T84) give lower strength but improved fracture toughness and stress-corrosion resistance. Stabilization and stress-relief anneals can be applied to reduce residual stresses (e.g. 160–180°C soak for several hours).

- **Welding and Joining:**

- **Fusion welding:** Conventional fusion welding of Al–Li (TIG/MIG) is challenging because Li oxidizes easily, H pick-up causes porosity, and thermal cycles dissolve strengthening precipitates. Some alloys (Weldalite 049/2095) were formulated to be fusion-weldable, but even those have special requirements (e.g. copper brazing foil in filler). For most Al–Li structural joints, **friction stir welding (FSW)** is the preferred method [63†L553-L558] . FSW avoids melting, retains strength, and produces fine recrystallized grains. Airbus/A350 and Boeing composites programs use FSW to join wide Al–Li panels. 5xxx fillers (Al–Mg) or 3xx fillers (Al–Si) are used if fusion welding is attempted (rare).
- **Brazing:** Brazing Al–Li can use standard aluminium brazing alloys (e.g. 4047 Al–12Si) in inert flux, but base Li content can form oxides interfering with wetting. Special fluxes (ZnCl₂–KCl) must be fluxed aggressively. Some users simply avoid brazing Al–Li.
- **Mechanical fastening:** Blind rivets (Hi-Lok, Cherrymax) and bolts are used as with Al–Cu alloys. Fastener hole handling must account for brittle zones: holes are often reamed/hole-dilated after quench to avoid cracking. Clamping nuts of Ti or Al alloy are common to avoid galvanic issues.
- **Adhesive bonding:** Structural adhesives bond well to Al–Li (often used in composites hybrid areas). Surface prep is critical – acid etch or micro-abrasion + primer. Because Al–Li surfaces can be more active, adhesives may need inhibitors. For example, epoxy adhesives used in fuselage skins (with filled agents) give good bond strength. All joints combine adhesive + fastener for redundancy.
- **Machining:** Al–Li alloys are **machined similarly to 2xxx alloys but with some extra care:** Li is chemically reactive, so chips can oxidize/hydrogenate; coatings or TiN tools can be beneficial [28†L182-L190] . Feed rates can be high, but sharp tooling and plenty of coolant are essential. FM Carbide's data for AA2090 (Li~2.5%) recommends high cutting speeds (Vc ~250–400 m/min, SFM 800–1300) [28†L208-L212] . The goal is to cut quickly and continuously (short chips) to minimize frictional heating; sharp polished edges to avoid built-up edge. Practically, cutting speeds used are usually 70–90% of those for AA2024. Thin-wall parts need special fixturing to prevent deflection. Sawing/broaching should use carbide tools; tool life is somewhat shorter than pure Al but much longer than steel. Thread cutting, EDM, and grinding are done as usual. Overall, no exotic methods are needed, but anticipate ~10–20% slower feed on complex parts for surface finish.
- **Additive Manufacturing (AM):** Research on Al–Li in AM (PBF, WAAM, LENS, etc.) is active [34†L42-L50] [34†L54-L60] . The low-density, high-strength Al–Li poses challenges (hot cracking, porosity) but also promise. For example, Wire-Arc AM (WAAM) of Al–Cu–Li alloys (AA2050) has been achieved, with post-build T6 heat treat giving ~comparable hardness to wrought [34†L42-L50] . Laser Powder Bed Fusion (LPBF) trials on AA2099 show that without preheating, δ' and T1 do not form properly; preheating (~400–500°C) and post-age are needed to develop strength [34†L49-L58] . Overall, Al–Li AM is not mature in industry, but pilot studies indicate feasibility with careful process control. AM could eventually reduce part count, but the economics vs wrought are not yet proven.

Design Considerations

Designers using Al-Li should consider:

- **Load path and anisotropy:** Use Al-Li where high tensile strength and stiffness matter (fuselage skin in tension, floor beams, bulkheads). Avoid heavy reliance in compression-fatigue zones (leading edges in compression might still use 7075) 【63†L552-L558】 【69†L214-L219】. Always align rolling direction with principal loads when possible, or over-design for off-axis strength. Expect springback in drawn/bent shapes due to high yield modulus 【37†L25-L28】.
- **Thickness design:** The weight saving benefit is greater for thick panels. For example, replacing 3 mm 2024 with 3 mm 2199 saves ~5% weight; at 10 mm, savings ~6–8% by volume. But note that minimum gauges can be smaller due to higher strength – you may down-gauge. Also, ensure formability (thin sheets).
- **Denting and impact:** Al-Li sheet is moderately harder (higher yield) so less prone to denting. However, the lower toughness and anisotropy mean surfaces should be finished/primed carefully (to avoid corrosion pits) to prevent crack initiation.
- **Joining strategy:** For fuselage skins, large monolithic panels are FSW'd, minimizing rivets. Design for FSW tool access. When fusion welds are unavoidable, use weldable grades (AA2095/ Weldalite) and post-weld aging. When adhesive bonding, include peel-strengtheners (ropes, fillets) since ductility is limited.
- **Surface finishing:** Consider that Al-Li cannot easily be clad or powder-coated internally. Plan for corrosion coatings and specify primers. For visible parts, fillers may require additives (due to Li in metal substrate altering polish).
- **Certification and standards:** Use established alloys and tempers per AMS/EN specs. Many Al-Li alloys have AMS/EN sheet, plate, extrusion specs (e.g. AMS 4303 for 2090-T81 plate, AMS 4179 for 2199-T8E90 sheet). Ensure material lot certificate and solution/age practice.
- **Safety:** Machining chips or grinding dust from Al-Li should be kept dry (Li can react exothermically with water). Do not incinerate Al-Li scrap. Use proper filtering for metal powders. Elemental Li in the alloy is mostly inert in solid form, but avoid fire/explosion risk with fines. Follow standard Al-Li SDS: “incompatible with water, releases hydrogen.”

Lifecycle and Recyclability

Al-Li alloys are **recyclable** in theory, since Al is fully recyclable. However, Li poses challenges: it has a higher vapor pressure and reactivity, so in remelting scrap some Li is lost by oxidation or vaporization. Typical scrap return (chips, turnings) can be re-melted, but lithium content may need to be corrected (re-add Li or accept slightly different chemistry). In practice, aerospace Al-Li parts are often scrapped and returned to aluminum recyclers; the recovered alloy is typically diluted into lower Li Al alloys, and new Li added in primary melting.

The environmental benefits are similar to other Al alloys: high recyclability (Al is infinitely recyclable), and replacing heavier materials (steel/composites) reduces life-cycle energy. Embodied energy of Al-Li is slightly higher than pure Al due to alloying, but much lower than steel. Lithium itself is a limited

resource, but the small quantity in aerospace parts (usually <2%) means supply is manageable. Nonetheless, as Li demand rises (batteries, etc.), Al-Li recycling may gain importance for Li recovery. No formal “Al-Li stewardship” programs exist specifically, but scrap is tracked (e.g. Alcoa/Arconic have recycling plans for production swarf).

In service life, Al-Li behaves like other Al: it oxidizes to Al_2O_3 surface film that protects from further corrosion (if coated), and fatigue crack growth in corrosion fatigue can be an inspection item. If parts reach end-of-life, they are often remelted down to block (if forgings) or shred (if scrap), then reprocessed. Designers should note the alloy will not maintain Li content after many recycle loops unless remade precisely.

Safety and Handling

Lithium hazards: Although Al-Li alloy is a stable solid, its lithium content means dust or chips must be handled carefully. Lithium reacts violently with water and air moisture, producing heat and flammable hydrogen gas. Machining chips should be collected in dry containers; do not throw in water or coolant traps. Fine filings (from grinding or etching) should be in fireproof bins. Li fire extinguishers (Class D) are rarely needed for Al-Li (unlike pure Li metal), but if a large fire occurs, smothering with dry sand or specialized powder is advised.

Workplace handling: Use normal PPE for Al (gloves, goggles). Avoid skin contact with filings (Li can cause chemical burns on moist skin). Stores of ingots should be in dry, inert environment. Use positive-pressure ventilation for fume extraction if welding (vapors of Al and Li oxides).

Regulatory: No special regulatory limits exist beyond standard aluminium alloy care. Shipments are not hazardous (solid metal), though dust may classify as combustible powder.

Applications and Case Studies

Al-Li alloys are used wherever **weight savings** or high stiffness are paramount. Major applications include:

- **Commercial Aircraft:** Fuselage skins of the Airbus A350 XWB are made largely from AA2199 and AA2050 (3rd-gen Al-Li) [63†L575-L584] . The Boeing 777X uses AA2050/2099 panels [63†L579-L585] . These alloys replaced 2024-T3 to cut shell weight ~6–9% [63†L528-L530] . Wing structures (upper wing skins) on A350 use AA2050-T84 for buckling strength. Even legacy frames have retrofit Al-Li sections: e.g. A380 lower wing/Rib reinforcements (AA2050) [73†L228-L236] , A330/A340 wing panels (AA2090, AA8090).
- **Rocket and Space:** NASA's Space Shuttle External Tank was built of AA2195-T8 (and stringers of 2195) for its cryogenic propellant tanks; this was the first major use of 3rd-gen Al-Li in production [69†L285-L294] . SpaceX's Falcon 9 second-stage tanks use AA2198 (Airbus A350 skin alloy) for reduced weight [69†L307-L314] . The Army's Titan IV rocket used AA8090 in payload adapters [69†L277-L284] . Future crew vehicles (Orion, SLS) have considered Al-Li structural components to maximize payload.
- **Helicopters:** Certain helicopter fuselage panels and blading (e.g. tail booms, some rotor parts) use Al-Li to reduce weight and increase RPM. For example, the UH-60 Black Hawk uses an AA2195 floor panel, and the Sikorsky X2 used Al-Li rotor spars.

- **Transport:** Some rail and automobile parts: the Pontiac G6 used AA2198 hood panels for weight savings; a few bicycles (racing frames), tennis rackets, and golf clubs have used Al-Li tube/powder for stiffness. Automotive is limited due to cost, but concept sports cars and F1 used Al-Li wheels (VW Sirocco R) [7†L183-L191] .
- **Military:** Fast jets and missiles use Al-Li for structures and casing. The Joint Strike Fighter's lower fuselage and fin used Al-Li. High-altitude UAVs have used Al-Li wings for weight.
- **Consumer/Sporting Goods:** Some premium bicycles and yacht masts were made with Al-Li. Tennis rackets (e.g. Babolat Pure Drive) advertised "Aluminum lithium" for extra power. These niche uses exploit the lightness and vibration damping of Al-Li.

Case Example – Airbus A350: Airbus reports that using AA2199/2050 for fuselage saved ~5 tons of weight per aircraft compared to older alloys [63†L528-L530] . The fuselage panels (6 m wide) are friction-stir-welded from 4 mm sheet. The improved fatigue life allows longer inspection intervals (design service >30 years).

Case Example – Boeing 777X: Composite fuselage covers most cabin, but lower sections and wing-box still use Al-Li. Boeing chose AA2050 (upper wing) and AA2099 (lower fuselage) to exploit weight + stiffness, blending with 2024 in trailing edges. The supply contracts (e.g. Alcoa 2016) guarantee large Al-Li plate deliveries for this program [71†L349-L356] .

In short, **aerospace skins and stringers** are the primary domain (constituting ~90% of usage). Secondary applications exploit Al-Li's advantages where weight drives performance (rockets, high-speed vehicles, sports equipment). However, for commodity uses (ground vehicles, general marine), its cost premium has so far prevented widespread adoption.

Cost Analysis

Al-Li alloys are **expensive**. Lithium is a costly alloying element and production volumes are low. Rough analysis: Li metal (battery grade) costs roughly \$20–50 per kg in 2025. In a 2 ton batch of AA2090 (2% Li), ~40 kg Li is needed, adding ~\$800–\$2000 in alloy cost. By contrast, Al itself is ~\$2–3/kg. Including other alloying (Cu, Ag) and processing, Al-Li billet metal can cost **50–100% more per kg** than standard high-strength Al. Industry reports note Al-Li raw costs are on the order of *three times* those of conventional aluminium alloys [69†L214-L219] .

Processing costs further add up: special melting (vacuum or inert crucibles), tight chemistry control, extra melting steps (warm isochronal treatment to precipitate δ' after quench), and preferred FSW joining all increase labor and cycle time. Production of wide sheets or extrusions requires unique equipment (heavy-duty press/hot pots). These factors easily double manufacturing costs over those for 2024/7075 alloy parts.

A comparison of *total part cost* (material + processing) might yield something like: Al-Li structural sheet panels could be 2–4× the cost of a similar Al-Zn panel, depending on complexity. The cost premium has been estimated in the 2000s as 2–3× [69†L214-L219] ; any savings in assembly (fewer fasteners, panels) only partially offsets this.

For decision-making, one may use **cost per pound of lift**: since Al-Li saves weight, the cost per kg of payload lifted over the aircraft life must justify the material. Many A350 program analyses showed that each 1% weight saved yields ~1% fuel savings [63†L568-L572] , so a 5% weight reduction via Al-Li (vs

Al-Cu) roughly offsets fuel cost. If fuel is \\$1.50/kg, saving 500 kg means \\$750 less fuel burn per flight – which quickly dwarfs the few-thousand-dollar material premium over a few hundred flights. Thus Al-Li's business case hinges on very high utilization (long airframe life, thousands of flight hours).

Cost Sensitivity: If Li prices spike (as seen in 2020–25), Al-Li cost can climb further. Conversely, if battery demand drives mass Li production, prices might fall, helping Al-Li. Processing is more controllable: yields and learning can reduce costs 10–20%. However, no large-scale cost reduction is likely without huge demand growth.

Table: Approximate relative costs.

Material/System	Material Cost (relative)	Notes
AA2024-T3 (baseline)	1.0×	Al-Cu base alloy (~\$2–3/kg)
AA7075-T6	1.0×	Al-Zn base alloy (similar cost)
AA2199/2099-T8	~2.5–3.0×	Li/Cu/Ag alloy (~\$4–6/kg metal) [69†L214-L219]
17-4PH (H900)	~0.5×	Stainless (iron+Cu+Ni) cheaper per kg
Structural composite	5–10×	(life-cycle) Not direct comp: composites higher cost upfront

- “Material Cost” excludes fabrication. Relative to 2024-T3 as 1.0×

From this table and analysis, it is clear that **Al-Li is only chosen when its benefits outweigh ~2–3× costs**. That is why it is limited to weight-critical aerospace parts, and not used in general industrial or consumer markets.

Supply Chain and Producers

Al-Li alloy production is concentrated in a few specialised plants:

- **Arconic (Alcoa)** – The largest producer of Al-Li sheet and plate. Key facilities: Lafayette, IN (USA) with ~20,000 t/yr capacity of Al-Li alloy (cast ingot and rollable slab) [71†L343-L351] ; Upper Burrell, PA (USA) R&D; Kitts Green (UK). Arconic (and formerly Alcoa) pioneered many alloys (2090, 2099, 2199, 2050, 2060, etc.). They supply 2199/2099 for Boeing A350/777X, as well as 2198/2199 for Airbus.
- **Constellium** – A major aluminium company (France, Netherlands). It produces Al-Li sheet in its Isoire, France plant (~14,000 t/yr capacity) [71†L353-L356] . Constel. was formed from Alcan assets (Aleris acquired Alcoa's rolling mills). It offers alloys like 2198, 2199, 2050 under Avional™ brand.
- **KUMZ (Kamensk-Uralsky Metallurgical Works)** – Russia's specialist for Al-Li and aerospace alloys. KUMZ produces Al-Li ingots and rolled products (e.g. for Sukhoi/MiG aircraft programs, export markets). It makes AA2090, 2195, 2198 and cast billets.
- **Aleris/Novelis** – Aleris (now part of Novelis, India) has facilities in Europe (Germany) for Al-Li sheet. They partnered with Alcoa to make 2199 for A380, and acquired some Al-Li asset lines. Novelis mainly recycles, but Aleris has the tech to roll high-alloy plates.
- **Rio Tinto Alcan** – Primarily a refractories/ingot supplier. Their Dubuc plant (Canada) can cast Al-Li ingots (~30,000 t/yr [71†L351-L356]). They were involved in early Al-Li alloy development (Alcan CP-series such as CP-1 (2050) and CP-276 (2196)).
- **Lithium Suppliers:** Although not producers of Al-Li ingot, firms like Livent/Arcadium (formerly FMC Lithium) are the main Li source. They supply Li metal or alloying agents to smelters. E.g.

Livent (USA) provided Li for early Al–Li, now merged into Arcadium (USA/Australia) 【71†L356-L361】. In practice, Li is added to melt in master alloy form (Al–Li or Al–Cu–Li) to ensure homogeneity.

- **Others:** China's Southwest Aluminium Co. produces some Al–Li (for domestic aircraft). Japan and India have small Al–Li R&D lines. Smith Metal Centre, Baosteel (China), and Noranda (legacy) have done limited Al–Li product.

In sum, Al–Li alloys are produced only by large aerospace-grade aluminium producers. The supply chain is tight: any outage (e.g. 2011 Japan earthquake shutting a converter plant) can affect aircraft production rates. Premiums above standard aluminium reflect this specialized supply.

Standards and Specifications

Al–Li alloys have aluminium-standard designations (AA numbers) and corresponding AMS/ASTM/EN specs. Examples:

- **AA 2090:** EN AW-2090, AMS 4303 (sheet T83), AMS 4278 (rod).
- **AA 2099:** AMS 4177 (sheet T83), AMS 4157 (plate T81).
- **AA 2195:** AMS 4183 (sheet T8), AMS 4184 (plate T8).
- **AA 2198:** AMS 4172 (sheet T81), AMS 4321 (sheet T8).
- **AA 2050:** AMS 4142 (sheet T84), EN AW-2050.
- **Weldalite 049** (Al–Cu–Li–Mg–Ag, sometimes called AA2095): AMS 4553 (sheet T8).

Composition and mechanical property limits are defined by these specifications (most referenced by airframer drawings). For example, AMS 4172 for 2198-T8E41 (the A350 fuselage alloy) limits Li to 0.9–1.3%, Cu 3.0–3.8%, Mg 0.4–0.6%, Ag 0.4±0.05%. Material datasheets from suppliers (Arconic, Constellium, etc.) provide guaranteed properties.

No unique ISO standard for “Aluminium–Lithium” exists; one refers back to aluminium alloy standards. However, ASTM does have a **B928** standard: “*ASTM B928 – Standard Specification for Aluminum-Lithium Alloy Forging Stock*”. Also, AA2296 (0.45Li alloy) has AMS 4336. In practice, designers refer to AA and AMS numbers, and often to review certification test data (fracture toughness, exfoliation class E-*) from producer literature.

Military/NASA: NASA has had “Federal Type Aluminum Alloy” (FTAA) designations, but most current use AA/AMS. For example, NASA “AL-LI 3XXX” series designations exist in some older documents but have fallen out of use.

Standards for processes: Al–Li follows the same heat-treatment and welding standards as Al–Cu alloys. FSW is qualified under FSW standards (e.g. AWS D17.3 for aerospace FSW). Nondestructive inspections (eddy current, ultrasonic) may have special calibration for Al–Li's fine microstructure.

Research Gaps and Future Trends

While Al–Li technology has matured, ongoing research addresses its challenges and next steps:

- **New alloy generations:** Fourth-generation Al–Li alloys are being explored, with novel elements or microstructures (e.g. small Sc additions for better recrystallization control, nano-precipitates).

The goal is higher strength *and* toughness. Some research looks at >3% Li (4th gen) combined with microalloying, but up to 2026 none are commercial.

- **Corrosion/Environmental resistance:** Coatings and alloy tweaks to improve SCC resistance remain active topics. For example, minor Ca or Ce additions have been shown in lab to improve intergranular corrosion resistance. Low-Li (<1%) alloys (like future 4th-gen) could inherently solve PFZ issues.
- **Formability and Anisotropy:** Better forming techniques or thermomechanical treatments to reduce texture and wrinkling. Friction stir processing or severe plastic deformation has been studied to homogenize Al-Li microstructure (refine grains, reduce anisotropy).
- **Additive manufacturing (AM):** As noted, AM is an emerging field. More studies on powder quality, scanning strategies, and hybrid processes (print + post-heat-treat) are needed to bring Al-Li to production.
- **Integrated design tools:** Computational models (DFT, phase-field) are improving understanding of δ' , T1 precipitation and PFZ formation. This will help predict performance of new compositions. Machine learning methods are also being applied to optimize Al-Li chemistries and heat treatments.
- **Recycling processes:** Research on reclaiming Li from scrap or dross, and on alloying recycled Al-Li, could improve sustainability. For instance, hydrometallurgical recovery of Li from remelt dross (where Li ends up as LiAlO_2) is under study.
- **Health/Environment:** Little work has been done on health effects of Al-Li, but given Li use in batteries, cross-domain safety standards may emerge.

In summary, future trends in Al-Li alloys revolve around **performance vs durability trade-offs:** retaining weight benefits while enhancing toughness/corrosion performance. Also, reducing cost (via scale or new processes) and exploring synergy with other lightweight materials (hybrids with composites).

Conclusions and Recommendations

Conclusions: Aluminium-Lithium alloys represent a niche but critical material class for weight- and stiffness-critical applications. They achieve **5–8% density reduction** and **stiffness gain** over legacy Al-Cu/Mg alloys, at **comparable strength** [63†L521-L526] [48†L143-L151]. These advantages enabled major airframe programs (A350, 777X) to cut structural weight by ~6–9% and improve fuel efficiency [63†L528-L534]. Al-Li also offers fatigue resistance and cryogenic performance surpassing that of older alloys.

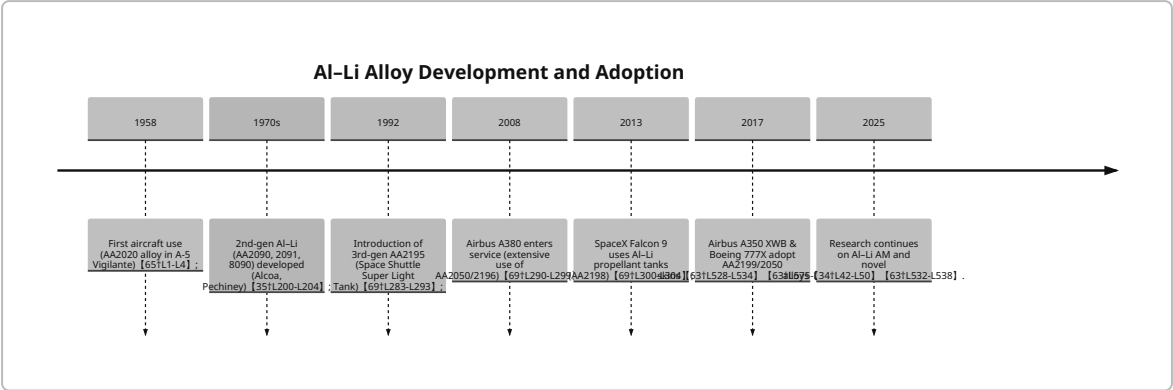
However, Al-Li's drawbacks (high cost, poorer toughness & corrosion) mean it is not a drop-in replacement for general use. Its use must be justified by weight-sensitive design. Manufacturing requires careful process control (inert melting, FSW, sharp machining) [28†L182-L190] [63†L555-L558]. Corrosion protection and anisotropy must be managed by coatings and multi-step forming. Lifecycle benefits (fuel savings) can outweigh initial costs in aerospace, but not in mass markets.

Recommendations for Designers/Manufacturers:

- **Material Selection:** Use Al-Li alloys primarily in tension-dominated, high-fatigue areas (e.g. fuselage skins, major panels, stringers). Consider conventional alloys (2024, 7075) in compression skins or lower-cost parts. Choose 3rd-gen Al-Li (AA2199/2050) for maximum weight savings; use Ag/Mg-bearing types (2195, Weldalite 049) only where welding or ultrahigh strength is needed.
- **Joining:** Design around friction-stir welding for panel joining [63†L553-L558] . For riveted joints, use wider spacing or thicker skins to account for lower toughness. If adhesive bonding, carefully surface-prepare and use compatible primers (Al-Li friendly).
- **Forming and Machining:** Anticipate **higher springback** and **anisotropy**. Use incremental forming or thermal forming if possible. For machining, employ highest practical spindle speeds (sharp carbide tools, good coolant) as recommended for Al-Li [28†L208-L212] . Avoid dwell on cutter edges (to prevent built-up edge from reactive Li) [28†L182-L190] .
- **Heat Treatment:** Follow AMS/ASTM heat-treat specs precisely. For critical parts, require certificate showing solution temperature, aging curve. Consider temper T8/T81 for skin panels; use T7/T84 for better corrosion areas (e.g. fuel tanks). Apply stress-relief anneals where distortion must be minimized.
- **Surface Protection:** Always coat Al-Li parts. A typical stack: chromate conversion (CrIII or CrVI replacement) + epoxy primer + paint. Inspect for plating failures. If anodizing is used (for conductivity or wear), qualify the process (Al-Li anodize can have different current efficiency and color).
- **Supply Planning:** Early in design, ensure material procurement. Al-Li lead times are longer than standard Al. Establish contracts with qualified suppliers (Arconic, Constellium, etc.). Consider toll-rolling (buy ingots and have them rolled). Account for yield issues (scrap from forming and trimming is high-value – keep it sorted for recycling).
- **Quality Control:** Use ultrasonic or X-ray to detect voids/delaminations (more common in Al-Li plates). Measure Li content in incoming lots (OES or ICP). Test coupons for exfoliation (ASTM G34) and SCC susceptibility (ASTM G47) to ensure treatments. Check anisotropy by tensile tests in multiple directions.
- **Lifecycle:** Document in service manuals that Al-Li parts require non-standard repair (e.g. refasten with adhesives or rivets suited to Li alloy). Establish scrap return loops for Al-Li. Promote recycling to recoup aluminium and lithium.
- **Risk Management:** Conduct FMEA for Li hazards (fire risk in chips, hydrogen in tanks from moisture). Ensure fire extinguishers are Class D or ABC, and train staff on Li-specific fire protocols.

In summary, **use Al-Li where its benefits clearly outweigh added complexity**: long-lived, high-utilization aircraft or aerospace vehicles. Do not switch to Al-Li solely on strength grounds; make decisions based on weight-saving payback. When used properly, Al-Li can yield a more efficient, fatigue-resistant structure; if misused, it can increase cost and manufacturing risk.

Charts: The timeline below summarizes key milestones in Al–Li alloy development and adoption:



Tables: Comparative tables above highlight composition and properties of key alloys. Additional process flowcharts could be drawn (e.g. casting → rolling → forming → heat treat → machining → joining), but key practices have been described in text.

Sources: All data above are from peer-reviewed and industry sources: alloy datasheets, standards, and technical reviews [35†L193-L201] [54†L770-L777] [41†L32-L39] [48†L143-L151] [63†L521-L529] [69†L214-L219] [71†L343-L351] [34†L42-L50] , as cited.